14.11.1 Description

In this hands-on lab, we will create a simple **Calculator API** using **Laravel 12**. The API will allow users to perform basic arithmetic operations (addition, subtraction, multiplication, and division). We will implement a **Data Transfer Object (DTO)** to handle request validation and data structuring.

14.11.2 Objectives

By the end of this lab, you will:

- Understand how to create a **REST API in Laravel 12**
- Learn how to use **Controllers and Routes** in Laravel
- Implement a **Data Transfer Object (DTO)** for request validation
- Apply **Service Layer Architecture** to separate business logic
- Test the API using **Postman or cURL**

14.11.3 Prerequisites

Before starting, ensure you have:

- **PHP 8.2+**, Composer, and MySQL installed
- **Laravel 12 installed**
- **Postman or cURL** for API testing
- Basic knowledge of Laravel

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

14.11.4 Steps

Here are the steps to complete this lab:

14.11.4.1 Step 1: Create a New Laravel 12 Project

Open your terminal and create a new Laravel project:

```
laravel new calculator-api
cd calculator-api
code .
```

You should see the Laravel project structure in your code editor.